

Supplementary Material S1_PRISMA

Systematic Review Protocol (PRISMA 2020)

Manuscript Title: HIMS CDI: A Governance Specification and Assurance Architecture for CTI-Driven Cybersecurity Risk Assessment in Healthcare

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DOI of main manuscript: [to be assigned]

Repository: <https://github.com/oluseun-akeredolu/HIMS-CDI>

Zenodo archive: <https://zenodo.org/records/16897633>

OSF protocol registry: <https://doi.org/10.17605/OSF.IO/3CW8Y>

1. Objective

The systematic review aimed to identify, appraise, and synthesise existing approaches to hybrid cyber threat intelligence (CTI) fusion in healthcare cybersecurity risk assessment (CSRA). The primary focus was on three governance-critical dimensions: semantic traceability (linking CTI artefacts to clinical risk constructs), regulatory embedding (clause-level compliance mapping), and deployment feasibility (readiness for resource-constrained clinical edge environments).

This document retrospectively summarises the methods that were planned and executed for the completed review; it is not a prospective protocol. All numbers and criteria are drawn from the final review records.

2. Eligibility Criteria

Studies were selected according to the criteria defined in the inclusion rubric (Supplementary Material Scoring_Rubric.pdf). In summary:

Inclusion criteria

- Empirical or technical evaluation of hybrid CTI fusion for cybersecurity risk assessment.
- Direct relevance to healthcare (validation on healthcare datasets, clinical scenarios, or explicit healthcare use cases).
- Published in Q1-ranked journals or peer-reviewed conference proceedings.
- Combined score ≥ 3 on the two-dimensional rubric (Empirical Evaluation & Methodological Rigour + Healthcare Validation Context).

Exclusion criteria

- Purely theoretical or non-empirical studies (e.g., generic cybersecurity guidelines, industry reports, news articles).
- Single-modality CTI approaches without any hybrid fusion.
- Studies lacking healthcare validation context (generic IT or non-clinical critical infrastructure).
- Duplicate publications or pre-peer-review manuscripts without accessible full texts.

The full exclusion log with individual reasons is provided in Supplementary Material Study_Exclusion_Log_69.csv.

3. Information Sources

The following electronic databases were searched:

- Scopus
- Web of Science Core Collection
- IEEE Xplore
- ACM Digital Library
- PubMed

The search was conducted on 25 October 2024, covering publications from 1 January 2020 to 25 October 2024. Additional records were identified through manual reference list screening; no automated tools were used.

4. Search Strategy

A representative search string (Scopus) is shown below. The complete database-adapted search strategies are documented in Search_Strategy_and_Criteria.pdf and archived in the OSF repository.

(TITLE-ABS-KEY ("cyber threat intelligence" OR "threat intelligence" OR CTI)

AND

TITLE-ABS-KEY ("risk assessment" OR "security assessment" OR CSRA)

AND

TITLE-ABS-KEY (healthcare OR "health care" OR medical OR clinical OR hospital))

AND PUBYEAR > 2019 AND PUBYEAR < 2025 AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp"))

Database-specific filters were applied where available (document types: Article, Conference Paper; language: English). No machine-learning classifiers were used in screening.

5. Selection Process

All retrieved records were imported into EndNote X9 and duplicates removed. Two independent reviewers (OA and a second experienced researcher) screened titles and abstracts against the eligibility criteria. Any record considered potentially relevant by at least one reviewer proceeded to full-text assessment.

The same two reviewers independently evaluated full-text reports using the standardised inclusion rubric (Scoring_Rubric.pdf). Disagreements were resolved through structured discussion; where agreement could not be reached, a third reviewer was consulted. A Delphi process involving a panel of NHS clinical safety officers and cybersecurity architects was employed to adjudicate borderline cases concerning healthcare specificity and deployment feasibility.

Pre-filtering restricted full-text assessment to studies from Q1-ranked journals or peer-reviewed proceedings.

6. Data Extraction

Data were extracted independently by the two reviewers using a standardised codebook (Extraction_Codebook.pdf). The extraction schema captured:

- Bibliographic information
- Methodological paradigm and fusion architecture
- Healthcare specificity (level 0–3)
- Regulatory embedding (level 0–3)
- Validation rigour (level 0–4)
- Operational readiness (level 0–3)
- Thematic codes (open, axial, selective) derived via grounded theory

Inter-rater reliability was monitored at key stages:

- Extraction calibration (training on 3 studies): $\kappa \geq 0.80$ required.
- Full extraction: Cohen's κ ranged from 0.72 to 0.79 across coding dimensions.
- Inclusion rubric application: $\kappa = 0.76$.

All disagreements were resolved by discussion or Delphi escalation. Complete extraction data are available in the project repository (OSF, GitHub, Zenodo).

7. Risk of Bias Assessment

Given the heterogeneity of study designs (technical frameworks, empirical evaluations, systematic reviews), a formal risk-of-bias tool such as QUADAS-2 or ROBIS was not applicable. Instead, study quality was characterised through the taxonomy dimensions:

- Validation rigour (0–4)
- Methodological transparency
- Operational readiness
- Regulatory embedding

No studies were excluded solely on the basis of quality scores, but lower-scoring studies were given proportionately less weight in the narrative synthesis.

Potential publication bias (the higher likelihood of publishing studies with positive CTI-fusion results) is acknowledged but was not formally assessed. The included body of evidence is drawn entirely from peer-reviewed, Q1-ranked outlets and conference proceedings, which provides a degree of quality assurance but may also reflect selective reporting.

8. Synthesis Methods

A quantitative meta-analysis was not feasible due to heterogeneity in outcome measures. A narrative synthesis was conducted, structured around:

1. **Taxonomic classification:** Each study was mapped onto the six-domain CSRA design taxonomy.

2. **Thematic analysis:** Grounded theory coding was applied to identify recurring themes and failure modes.
3. **Failure mode consolidation:** The three dominant failure modes (semantic-pragmatic misalignment, regulatory-methodological drift, deployment discontinuity) were derived by axial coding of study limitations and validated through the NHS Delphi panel.
4. **Blueprint development:** The conceptual governance blueprint (HIMS-CDI) was synthesised from the taxonomy and failure-mode analysis.

No subgroup or sensitivity analyses beyond the Q1-only stability check (Supplementary Material sensitivity_analysis.csv) were pre-planned.

9. PRISMA 2020 Flow Diagram

Figure S1–PRISMA.1 – PRISMA 2020 flow diagram for the systematic review.

Stage	Number of records
Identification	
Records identified through database searching	427
Records identified through other sources (manual reference list screening)	0
Total records identified	427
Duplicates removed	(53)
Screening	
Records after duplicates removed (titles/abstracts screened)	374
Records excluded (title/abstract not meeting eligibility)	(262)
Eligibility	
Full-text articles assessed for eligibility	112
Full-text articles excluded, with reasons	(69)
<i>Lacked healthcare validation context</i>	<i>(31)</i>
<i>Used non-hybrid CTI fusion (single-modality)</i>	<i>(22)</i>
<i>Non-empirical (e.g., reviews, reports, guidelines)</i>	<i>(16)</i>
Included	
Studies included in qualitative synthesis	43

Note: The 69 excluded studies are documented in Study_Exclusion_Log_69.csv. The 43 included studies are listed in studies.csv.

10. PRISMA 2020 Checklist (Abridged)

Section/Topic	Item	Reported in
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Title	1	Main manuscript (Abstract)
Abstract	2	Main manuscript (Abstract)
Rationale	3	Main manuscript §1.2
Objectives	4	§1
Eligibility criteria	5	§2
Information sources	6	§3
Search strategy	7	§4
Selection process	8	§5
Data collection process	9	§6
Data items	10	Extraction codebook
Study risk of bias assessment	11	§7
Effect measures	12	Not applicable (narrative synthesis)
Synthesis methods	13	§8
Reporting bias assessment	14	Acknowledged in §7; see also main manuscript §5.7
Certainty assessment	15	Discussed in main manuscript §5.7 (Limitations)
Study selection	16	§9 (flow diagram)
Study characteristics	17	studies.csv, main manuscript §2
Risk of bias in studies	18	§7
Results of individual studies	19	performance_metrics.csv etc.
Results of syntheses	20	Main manuscript §2, §3
Reporting biases	21	Not assessed; acknowledged in §7
Certainty of evidence	22	Discussed in main manuscript §5.7
Interpretation	23	Main manuscript §5 (Discussion), §5.7 (Limitations)
Registration and protocol	24	OSF: https://doi.org/10.17605/OSF.IO/3CW8Y
Funding	25	Declarations in main manuscript
Competing interests	26	Declarations in main manuscript
Availability of data, code, and other materials	27	§11, main manuscript Data Availability

11. Data Availability

All extracted data, the full extraction codebook, inclusion/exclusion logs, and detailed PRISMA documentation are openly available in the following repositories:

- OSF: <https://doi.org/10.17605/OSF.IO/3CW8Y>
- GitHub: <https://github.com/oluseun-akeredolu/HIMS-CDI> (directory /review_artefacts)
- Zenodo: <https://doi.org/10.5281/zenodo.16897634>

The files included are:

- studies.csv – Master list of 43 included studies
- thematic_coding.csv – Qualitative coding data
- performance_metrics.csv – Quantitative metrics and deployment characteristics
- methodological_breakdown.csv – Paradigm analysis
- validation_rigour.csv – Validation methods and operational readiness
- regulatory_analysis.csv – Regulatory compliance mechanisms
- sensitivity_analysis.csv – Sensitivity analysis
- Study_Exclusion_Log_69.csv – Full list of excluded studies with reasons
- Search_Strategy_and_Criteria.pdf – Inclusion rubric and search strategy
- Extraction_Codebook.pdf – Complete codebook

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